Clinical Risk Assessment Report: Patient 90

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Borderline / Low Confidence
- Absolute Predicted Risk: 23.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.07x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +4.3%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

Clinical Sensitivity Analysis (Borderline Patient):

PREDICTION INSTABILITY WARNING: This patient's risk profile sits precisely on the borderline. Consequently, the model's prediction is highly sensitive. Minor biological variations or measurement errors in the following features would flip the final clinical prediction:

- Long Run High Gray-Level Emphasis (PET): An increase would alter the prediction outcome.
- Large Zone Low Gray-Level Emphasis (CT): A decrease would alter the prediction outcome.
- Tumor Pixel Correlation (CT): An increase would alter the prediction outcome.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 0.87 [Top 11%] (+4.3%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: 6.37 [Top <1%] (+2.0%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 0.77 [Top 22%] (+1.8%)**

Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.62 [Avg (65th %ile)] (+1.6%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.25 [Bottom 5%] (+1.6%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.76 [Top 22%] (+1.5%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

3. Protective / Mitigating Factors (Reducing Risk)

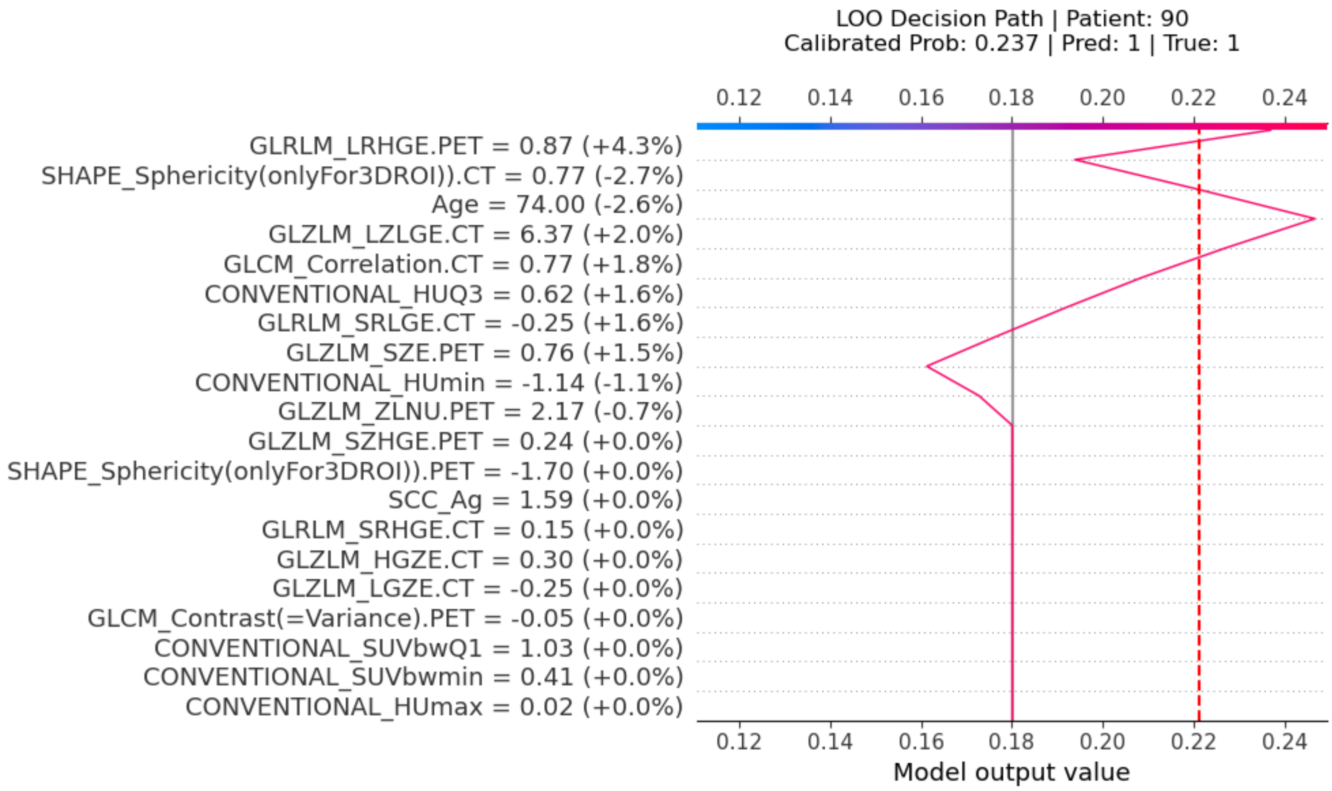
Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 0.77 [Top 21%] (-2.7%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Patient Age (Age) **Value: 74.0 [Top 13%] (-2.6%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.